# CJ Nour

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## Technical Skills

- Java
- Electrical Design
- Python
- PLC Programming
- React.js
- Git Source Control
- Testing & Commissioning Software
- Object Oriented Programming
- Data Structures & Algorithms

## Professional Experience

### Husky Technologies

*Electrical Design Engineer* — Jun. 2025 – Present

- Leading the electrical design of prototypes for a new machine generation. Creating schematics in **EPLAN**, CAD in **Siemens NX**, sizing cable assemblies, and validating grounding to **UL 508A**. Diagnosed and resolved a 9 year-old service issue related to Ethernet cable routing, **cutting critical communications faults by 100%**.
- **Resolved 30+ non-conformances** on a new machine family (UL & EN builds; CE and non-CE cabinets). Corrected panel schematic errors critical to safety devices, fixed assembly drawing errors. Achieved full release compliance by **eliminating 100% of the machine's electrical cabinets release blockers**.
- **Authoured and published three enterprise-level electrical standards** to drive proactive refresh of data and reduce engineering errors. Presented to engineering leadership to secure adoption across disciplines to actively **sustain a multi-million dollar per year business**.

### Tesla

*Controls Software Engineer Intern* — Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and developed a custom **TwinCAT HMI dashboard** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Contributed to commissioning **3 automation** machines by leveraging **Beckhoff TwinCAT3 Structured Text** to program critical components, including **VFDs**, sensors, and robotic and pneumatic devices. Debugged field I/O networks using multiple industrial communication protocols (**TCP/IP, EtherCAT, Profinet**).
- Led multiple **R&D** initiatives to solve critical engineering challenges to enhance **Beckhoff PLC** and **FANUC robot** performance by experimenting with **computer vision systems**, **time-of-flight lasers**, **profilometers**. Authoured **test result reports** and applied technical specifications to optimize device integration.

*Material Planner Intern* — Sep. 2023 – Dec. 2023

- Developed and sustained 3 **Excel VBA** tools to optimize inventory and logistics, **eliminating duplicate shipments by 100%**. Automated the parts reordering process, automated gap analyses for project needs, and streamlined shipping with auto-generated forms and emails - enhancing database operational efficiency.
- Leveraged these tools to **identify inefficiencies** in the existing third-party inventory system and built a business case to influence executives to create an internal software team to develop an improved, proprietary system.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

## Professional Experience Continued

### Moneris

*Technology Governance Intern* — May 2023 – Aug. 2023

- Reduced the application database **by over 50%** through **cost center analysis** and a detailed **general ledger audit**. Retired outdated applications and underutilized software licenses, resulting in **significant cost savings**.
- Orchestrated collaboration across departments to gather, validate, and standardize data for **cloud migration planning**, ensuring a seamless transition phase and enhancing application scalability.
- Strengthened IT governance and compliance frameworks by aligning the application inventory with corporate standards to streamline software usage across teams, resulting in a **reduction of licensing costs by over 30%**.

*Software Engineer Intern* — Jun. 2022 – Apr. 2023

- Developed a tool **boosting productivity for 30+ employees** by enabling credit card transaction analytics. Built with **JavaScript** and **React**, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted on **Parse**, optimizing personal and team tools by implementing reducers, utilizing dispatch, and **querying user database** properties.
- Collaborated in **agile** feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

## Projects

### Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection**, robotic arms designed in **CAD**, and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

### Autonomous RC Car

- Implemented self-driving algorithms coded in **C++ and Python** on an RC car to navigate through obstacles via **LiDAR** and **ultrasonic sensor** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **agile** team meetings as the Scrum Master to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

### Automated Vanity System

- Designed and fabricated a custom automated vanity system from a recycled oil barrel. Cut and welded a panel from the barrel to a recycled chair and added wheels for smooth accessibility.
- Engineered a motorized mirror and a **Bluetooth** sound system by wiring a recycled car window regulator motor to a momentary rocker switch and a power supply with an **AC/DC rectifier**. Researched **torque-speed characteristics** and motor specifications to ensure the motor could support the load with the supplied power.

## Education

**B. Eng., Electrical Engineering** — Graduated Apr. 2025

*McMaster University* — Hamilton, ON